AHMED ABDELMONEM MOHAMED ABDLGAYOUM

MECHATRONICS ENGINEER

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PROFESSIONAL SUMMARY

Mechatronics Engineering graduate specializing in embedded systems, robotics, and industrial automation. Skilled in microcontroller-based system design (Arduino, ESP32/ESP8266, STM32, Raspberry Pi), PLC-based process control, and deploying computer-vision and deep-learning models on low-power edge devices without cloud dependency. Hands-on experience across the full development cycle: mechanical design in SolidWorks, embedded C/C++ firmware, sensor integration, and industrial-panel electrical work. Seeking an entry-level engineering role in mechatronics, robotics, or automation.

WORK EXPERIENCE

Electrical & Instrumentation Engineer Jul 2025 – Oct 2025

HI TECHNICS SYSTEM SDN BHD | Cyberjaya, Malaysia

- Carried out electrical testing, panel wiring, and instrumentation activities in line with industrial safety standards, minimizing downtime through effective troubleshooting.
- Performed insulation resistance and earth continuity testing, and assisted with sensor installation and documentation — improving system reliability in industrial settings.

PROJECTS

Real-Time Embedded YOLOv11-NCNN Vision System for Autonomous Navigation

Designed and implemented an embedded AI vision system for autonomous navigation using Raspberry Pi 5 with CPU-only inference. Converted the YOLOv11 model to ONNX and optimized with NCNN for real-time edge deployment without GPU or cloud dependency. Detects traffic signs and integrates perception with decision-making and motor control.

Tech: Raspberry Pi 5, YOLOv11, ONNX, NCNN, Python, OpenCV

Smart Autonomous Lawn Mower Robot

Designed an autonomous robotic system using ESP32 for motor control and Raspberry Pi for vision processing. Implemented a comparative study between YOLOv8 and MobileNetV2 to improve grass-detection accuracy and navigation efficiency.

Tech: ESP32, Raspberry Pi, YOLOv8, MobileNetV2, OpenCV, Python, DC Motors

Pickup Robot using ROS2 & Deep Learning (Simulation)

Developed a simulated autonomous mobile robot with a 6-DOF robotic arm for intelligent object pickup and waste sorting. Integrated YOLOv8-based object detection with camera and LiDAR sensors in a Gazebo simulation environment.

Tech: ROS2, Gazebo, SolidWorks, Ubuntu, YOLOv8, LiDAR, 6-DOF Arm

PLC-Based Filling Tank System (Factory I/O Simulation)

Built a PLC-controlled automated filling tank system using Factory I/O simulation, performing sequential control for liquid filling, level detection, and pump operation via ladder logic, with a focus on industrial process sequencing.

Tech: PLC, Ladder Logic, Factory I/O

PLC-Based Warehouse Storage System (Factory I/O Simulation)

Designed and implemented a PLC-based warehouse automation system using Factory I/O simulation, managing automated storage and retrieval operations via conveyor systems, sensors, and sequential control logic.

Tech: PLC, Ladder Logic, Factory I/O

Fully Automated Soap Production System

Developed an industrial-style automation system controlling pumps, motors, and temperature regulation for soap production, implementing multi-stage process control across Arduino Mega and ESP32.

Tech: Arduino Mega, ESP32, DHT11, Level Sensors, Pumps, Motors

Fume Hood Monitoring System (IoT-Based Safety System)

Built an IoT-based environmental monitoring system using ESP32 for real-time gas, temperature, and humidity detection, with automated fan control and visual/auditory safety alerts.
Tech: ESP32, DHT11, MQ-5, Ultrasonic Sensor, LCD, IoT

SKILLS

Programming: C++, Python, Embedded C
Hardware / Microcontrollers: Arduino, ESP32, ESP8266, STM32, Raspberry Pi
Robotics / Vision: ROS2, OpenCV, YOLO
Automation: PLC / Ladder Logic, Factory I/O, IoT
CAD / EDA: SolidWorks, Fusion 360, Proteus, EasyEDA
Soft Skills: Problem Solving, Team Collaboration, Technical Documentation, Adaptability, Time Management

EDUCATION

Bachelor of Mechatronics Engineering (Hons.) Oct 2022 – Jul 2026
Universiti Malaysia Pahang Al-Sultan Abdullah (UMPSA)

CERTIFICATIONS

The Complete Electronics Course: Analog Hardware Design Udemy, Jan 2026

LANGUAGES

Arabic (Native) • English (Fluent) • German (Basic)

REFERENCES

Mr. Khairul Fikri Bin Muhammad — Supervisor
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